Problem A. Attacks Number

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

In the pre-season NBA game, the referees decided to stop the game as little as possible, and the teams treat each other very politely: the game has only training value, so each attack can result in either 0 points, or 2 points, or 3 points — the teams do not receive any one-point free throws. After one team finishes their attack, the other team starts their attack.

We will call an attack successful if the team scores a non-zero number of points. You are given the current score of the game. Find the minimum **total** number of successful attacks required to equalize the score in the game.

Input

The input file contains the score — one line consisting of the number of points scored by the home team and the number of points scored by the away team, separated by a space. The number of points for each team is a positive integer from 1 to 120.

Output

Output a single integer — the minimum number of successful attacks required to equalize the score in the game.

Examples

standard input	standard output
108 100	3
10 10	0
90 94	2

Problem B. Brands

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

One big company decided to change one of their brands. The brand shall be similar to existing one, but it shall be *readable*. The word is considered readable if it does not contains two vowels nor two consonants in the row. The management of the company considers the letters `aeiouy` as vowels, and other English letters as the consonants.

Given the current name of the brand, insert the minimal possible number of letters to make the name readable. If there are multiple solutions, find one that gives the lexicorgraphically minimal name.

Input

The first line of the input contains the current name of the team — the non-empty string of length no greater than 10^4 lowercase English letters.

Output

Print one line — the new name of the brand after the transformation.

Examples

standard input	standard output
ct	cat
rat	rat

Problem C. Classify the Roman Numerals

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Your task is to determine the number of symmetry axis of the given Roman Numeral.

The following table from Wikipedia displays how the Roman Numerals are written:

Individual decimal places	Thousands	Hundreds	Tens	Units
1	M	C	X	I
2	MM	CC	XX	II
3	MMM	CCC	XXX	III
4		CD	XL	IV
5		D	L	V
6		DC	LX	VI
7		DCC	LXX	VII
8		DCCC	LXXX	VIII
9		CM	XC	IX

Note that:

- The numerals for 4, 9, 40, 90, 400 and 900 are written using "subtractive notation" where the first symbol is subtracted from the larger one (for example, for 40 ("`XL`") '`X`' (10) is subtracted from '`L`' (50)). These are **the only** subtractive forms in standard use.
- A number containing several decimal digits is built by appending the Roman numeral equivalent for each, from highest to lowest.
- Any missing place (represented by a zero in the place-value equivalent) is omitted.
- The largest number that can be represented in the Roman notation is 3 999 (MMMCMXCIX).
- The letters `I`, `V`, `X`, `M`, and only those, are vertically symmetric, the letters `I`, `X`, `C`, `D`, and only those, are horizontally symmetric.

For example, 200 have the horizontal symmetry only (because `CC` have the vertical axis, but does not have horizontal one), 2000 have the vertical symmetry only(`MM` have the vertical axis of symmetry, placed between the letters, but does not have horizontal one), the number 1001 (`MI`) have no axis of symmetry at all, and the number 30 (`XXX`) have both axis (vertical one is placed in the middle of the second `X` letter).

Input

The first line of the input file contains a single integer T - the number of Roman numbers in the test ($1 \le T < 4000$).

Each of the following T lines contains a single integer: the decimal representation of the Roman Numeral. The integers are between 1 and 3999, inclusively.

Output

For each numeral, output `-` if the Roman numeral on it has horizontal symmetry, `|` if it has vertical symmetry, `+` if it has both, and `?` if it has neither.

Example

standard input	standard output
4 200 2000 1001 30	- | ? +

Problem D. Decimal-Friendly Numbers

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Let's consider the set S_N consisting of all positive integers whose decimal representation consists of exactly N digits and does not contain leading zeros. We call a number "decimal-friendly" if all its digits have a common divisor with 10, greater than 1.

Suppose we randomly and uniformly select K numbers from S_N. What is the minimum value of K such that the probability of having at least one decimal-friendly number among the selected numbers is greater than $0.999\,999\,999$?

Input

The input contains a single integer N ($1 \le N \le 50$).

Output

Output a single integer K — the minimum amount of numbers that need to be selected in order to have a probability of having a decimal-friendly number among selected ones greater than or equal to $0.999\,999\,999$. If the relative error of the answer does not exceed 10^{-6}, the answer is considered correct.

Examples

standard input	standard output
1	26
9	2211

Problem E. Escalators

Input file:	standard input
Output file:	standard output
Time limit:	2 seconds
Memory limit:	1024 megabytes

The new maze can be represented as a rectangular table with N rows and M columns, with an horizontal escalator located in each unit cell. If a toutist enters a cell, the escalator moves them to an adjacent cell in the direction of the escalator (one of the four neighboring cells). If the new cell is outside the labyrinth, the visitor exits the labyrinth.

To test the maze, Q independent experiments are conducted. In each experiment, a specific (exactly one) escalator is chosen and its direction of movement is changed. The starting cell for the visitor is also specified. It is required to determine whether the visitor will eventually exit the labyrinth using only the escalators, and if so, how many escalators they will use.

Input

The first line of the input contains two positive integers N and M ($1 \le N, M \le 1500$) - the number of rows and columns in the labyrinth.

Each of the following N lines contains M characters — the directions of the escalators. The direction is represented by one of the four characters '<', '>', 'v', and '^', which indicate the left, right, down, and up directions of the escalators, respectively. The upper left cell is located at the intersection of the first row and the first column.

The next line contains a single integer Q ($1 \le Q \le 10^5$) - the number of queries. Each of the following Q lines contains four integers S_x, S_y, C_x, C_y - the coordinates of the visitor and the coordinates of the escalator to be modified, respectively ($1 \le S_x, C_x \le N$, $1 \le S_y, C_y \le M$), and the character C_t specifying the new direction of the modified escalator.

Output

For each query, output the number of different escalators (including the initial one!) that the visitor will pass through if they can exit the labyrinth by moving only with the help of the escalators, or 0 if they cannot exit.

Example

standard input	standard output
3 4 >>v> ><v> ^>>^ 4 2 1 1 1 ^ 2 1 2 2 > 2 1 2 1 < 1 1 2 4 ^	0 6 1 8

Problem F. Fence Painting

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

The fence consists of N vertical boards. You have K different colors of paint, numbered consecutively from 1 to K. Some boards are already painted, while the rest are unpainted. You want to paint the fence in such a way that no two adjacent boards are painted the same color, without repainting the already painted boards. How many ways can this be done?

Since the answer can be very large, output the remainder when divided by 998 244 353.

Input

The first line of the input contains two integers N and K: the number of boards in the fence and the number of colors, respectively ($1 \le N \le 500\,000$, $3 \le K \le 10^9$).

Next, there are N integers from 0 to K — the boards in the fence, listed from left to right. If the corresponding number is 0, the board is unpainted, if it is positive — the board is painted in the corresponding color. It is guaranteed that if two painted boards are adjacent, they are painted in different colors.

Output

Output a single integer — the number of different ways to paint the boards according to the problem statement modulo 998 244 353.

Examples

standard input	standard output
4 3 1 0 2 0	2
3 5 3 0 3	4
10 123 2 0 2 3 1 0 2 1 15 0	1800964

Problem G. Generic Strings

Input file:	standard input
Output file:	standard output
Time limit:	2 seconds
Memory limit:	1024 megabytes

A string S is given, as well as x of its substrings (specified by the indices of their first and last elements; the letters in the string are numbered starting from 1).

The StringGPT string recognition system defines a string as *generic* if there are no two identical letters among the letters present in each of the selected substrings.

Over time, the letters from the string fall off in a certain order, leaving spaces behind. According to the StringGPT definition, spaces are not recognized as letters, so when determining if the string is generic, StringGPT only considers the letters that remain (in particular, after all the letters have fallen off, StringGPT will consider the resulting empty space as a generic string).

Given the string S, a set of x control substrings, and the order in which the letters disappear, determine after which disappearance StringGPT will consider the string as generic for the first time.

Input

The first line of the input contains a string S, composed of lowercase English letters ($1 \le |S| \le 10^5$).

The second line of the input contains a single integer x ($1 \le x \le 10^5$) - the number of control substrings. Each of the following Q lines contains two integers a_i and b_i ($1 \le a_i \le b_i \le |S|$) - the indices of the first and last characters of the current substring.

The last line contains $|S|$ pairwise distinct integers d_i ($1 \le d_i \le |S|$) - the indices of the disappearing letters, listed in the order of their disappearance.

Output

Output a single integer — the number of letter disappearances before StringGPT considers the string as generic for the first time.

Examples

standard input	standard output
qqqqq 2 1 2 4 5 2 4 1 5 3	2
pbbpbppb 3 1 3 4 7 3 5 6 3 5 1 4 2 7 8	5
nerc 1 1 4 1 2 3 4	0

Problem H. How Many Representations?

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

In the monetary system of Byteland, the denominations of all coins are elements of the sequence C_i, defined as follows:$C_1 = 1$, $C_2 = 2$, $C_i = C_{i-1} + C_{i-2}$ for all $i > 2$. Coins with the same denominations are considered identical, while coins with different denominations are considered distinct.

You need to find the number of ways to pay the sum of N using **distinct** coins. Two ways are considered different if a coin is present in one partition and absent in the other.

Input

The input contains a single integer N ($1 \le N \le 10^{18}$) - the amount of money to be paid.

Output

Output a single integer — the answer to the problem.

Examples

standard input	standard output
21	4
116	8

Note

For the first example ($N = 21$), possible exchanges: 21, 13+8, 13+5+3, 13+5+2+1.

Problem I. Interactive Permutations Strike Back

Input file:	`standard input`
Output file:	`standard output`
Time limit:	5 seconds
Memory limit:	1024 megabytes

This is an interactive problem

The jury has chosen a permutation of length N. You are given the number N and the number of inversions in the permutation. A pair of numbers forms an inversion if the larger number appears to the right of the smaller number. For example, in the permutation $(2, 1, 5, 3, 4)$, there are three pairs of numbers that form inversions: (2,1), (5,3), and (5,4). Therefore, the number of inversions in this permutation is 3.

You can ask the jury's program to swap the numbers at certain positions in the permutation. In response, the permutation will change, and you will receive the number of inversions in the new permutation. Your task is to guess the **initial** permutation in no more than $2 \cdot 10^5$ queries.

Interaction Protocol

The interaction starts with the jury's program printing two integers $1 \le N \le 10^5$ — the length of the permutation and $0 \le I \le 5\,000\,050\,000$ - the number of inversions in the initial permutation.

Then, your program can ask questions. Each question has the form `?` i j, where $1 \le i, j \le N$. After that, the jury's program swaps the i-th and j-th elements of the current permutation and outputs the number of inversions in the new permutation (the next query will be applied to the new permutation). You are allowed to ask no more than $2 \cdot 10^5$ questions in total.

If your program is ready to guess the permutation, output `!`, followed by a single line containing N integers from 1 to N: the initial permutation chosen by the jury's program. This action **does not** count as a question.

Please do not forget to print end-of-line character at the end of each question and answer printing, and flush the output after it using the `flush` function of your programming language. Otherwise you may receive the Idleness Limit Exceed (or Wall-Time-Limit Exceed) error.

Example

standard input	standard output
5 6 5 6 5 6	 ? 1 2 ? 1 2 ? 3 2 ? 3 2 ! 4 3 1 5 2

Problem J. Jungle And Trees

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

The jungle can be represented as the square with coordinates of vertices $(1, 1)$, $(1, N)$, (N, N) and $(N, 1)$. At each of N^2 integer points the tree is planted.

We consider the set of trees controlling, if the following property is held: for each tree in the set there are **no** other trees in the set that belongs to the straight line, connecting this tree with the point $(0, 0)$.

Given the N, find the maximal number of the trees in the controlling set.

Input

The input contains an integer N ($1 \le N \le 10^6$).

Output

Output a single integer — the maximum number of the trees in the controlling set.

Examples

standard input	standard output
6	23
42	1083

Problem K. King's Cup

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

For The King's Cup in Winter Sports, a new biathlon race format has been proposed.

The athlete starts from the base at point $(0, 0)$ and must pass through N shooting ranges (where N is even). The rifle can hold ammunition for only two shooting ranges (and can only be reloaded after firing all the rounds), so after every two shooting ranges, the athlete returns to the base to get ammunition for the next two ranges — or for the finish line if all ranges have been completed. Between shooting ranges and between the base and a shooting range, the biathlete can only travel in a straight line connecting these two points. The only point that the biathlete is allowed to pass more than once is the base (which also means that no two segments of the path can intersect at internal points and that the biathlete must enter and exit each shooting range exactly once).

Given the coordinates of the shooting ranges, determine the minimum length of the route that the biathlete must take from the start to the finish.

Input

The first line of the input contains a single integer N — the number of shooting ranges ($2 \le N \le 500$, N is even).

Each of the following N lines contains two integers x_i and y_i — the coordinates of the corresponding shooting range ($-10^6 \le x_i, y_i \le 10^6$).

It is guaranteed that all key points (shooting ranges and the base) are distinct and no three of them lie on the same line.

Output

Output a single real number — the minimum length of the route that the biathlete must take from the start to the finish, with an absolute or relative error no worse than 10^{-6}.

Example

standard input	standard output
4 -1 1 -1 4 1 1 1 4	17.074638376

Problem L. Long Sequence

Input file:	`standard input`
Output file:	`standard output`
Time limit:	10 seconds
Memory limit:	1024 megabytes

Given a sequence $A_1, \ldots, A_n$ of length n, and m operations, each operation modifies an A_i to k.

Before the first modification and after each modification, you are required to find a monotonic non-decreasing sequence $B_1, \ldots, B_n$ of the same length n, such that $\sum_{i=1}^{n}(A_i - B_i)^2$ is the minimum, and output the minimum value.

It should be noted that the impact of each operation is independent, that is, each operation will only affect the current query. In order to avoid precision problems, we ensure that this minimum value is a fraction, that is, it can be expressed as the division of two non-negative integers: x/y. Then you will output the value of $(x \times y^{P-2}) \bmod P$, which represents the value of x/y, where $P = 998\,244\,353$.

Input

The first line of the input contains two non-negative integers n and m, representing the sequence length and operands ($3 \le n \le 10^5$, $0 \le m \le 10^5$).

The second line has n positive integers separated by spaces, representing the sequence $A_1, \ldots, A_n$ ($1 \le A_i \le 10^9$).

Each of the following m lines contains two positive integers i and k, denoting that A_i is modified to k ($1 \le i \le n$, $1 \le k \le 10^9$).

Output

Output $m + 1$ lines with an integer in each line. The i-th line contains the answer after the $i - 1$-th modification. In particular, line 1 should be the answer to the initial position.

Example

standard input	standard output
5 3 10 3 5 7 5 1 2 1 5 5 7	28 2 4 26